

Workbook

Phase A. Frame

Name(s)
The case
Started

How this workbook works

Left page tells you, right page is yours

Every step has an instruction page on the left and a working page on the right. Everything needed to fill the right-hand page is on the left-hand page facing it, so nothing has to be remembered from an earlier step or looked up elsewhere. Read the left page, then write on the right.

The work on each right-hand page is set out as numbered exercises. Do them in the order they are printed.

Moving forwards is checked, backwards is always free

Each step ends with a **critical check**: a short list the participant checks themselves. The step is not finished until every line can be ticked off. Going back, on the other hand, costs nothing and is expected. When a later step shows that an earlier one was wrong, go back, change the earlier work, and write the date and the reason in the strip at the foot of that step's page. Nothing is rubbed out. What was first thought is evidence of where the thinking moved.

Two colours of pen make that visible. Use one colour for everything written the first time, and a second colour for every revision. A page in one colour has either been right first time or has not been revisited, and both are worth noticing.

Nothing here is assessed by filling it in

A full page is not the point. A page with four actors on it and a good reason for the other eight being left out is a better answer than twelve names.

Where the words are defined

The glossary on page 6 gives the short definition of every term this process uses precisely. The longer definition, with what the term excludes and why, is on the instruction page of the step that first uses the term.

The notes pages

The notes pages at the end, from page 26, are for anything that does not belong on a working page: questions that could not be answered, sources to find later, and disagreements the group did not settle. They are part of the work rather than a margin.

The thirteen steps

Five phases. This workbook covers **Phase A**, which is Steps 1 to 3. The later phases are added to this workbook as they are written.

A Frame

1	Define the problem, the desired change, and your position	9
2	Set the boundary and the perspectives	13
3	Describe behaviour over time	21

B Understand

4	Identify enablers and inhibitors	<i>later</i>
5	Map the causal structure	<i>later</i>
6	Analyse the mapping and write its narrative	<i>later</i>

C Choose

7	Identify leverage	<i>later</i>
8	Weigh consequences and interests	<i>later</i>
9	State the intervention hypothesis	<i>later</i>

D Test and plan

10	Test the hypothesis	<i>later</i>
11	Plan monitoring and learning	<i>later</i>

E Apply

12	Apply the analysis	<i>later</i>
13	Communicate to audiences	<i>later</i>

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What systems thinking is

A way of explaining why a situation keeps producing the same result, by looking at how its parts are connected rather than at the parts one at a time.

The starting claim

Most stubborn problems are not caused by one thing going wrong. They are produced, over and over, by an arrangement that is working exactly as it is arranged. Meadows puts the claim at its strongest: a system, to a large extent, causes its own behaviour (Meadows, 2008, p. 2). An outside event may set the behaviour off, but the same event applied to a differently arranged system produces a different result.

That is why a problem can survive several sincere attempts to solve it. Each attempt acts on a part. The arrangement that produces the problem is left intact, and the arrangement reasserts itself.

What follows from the claim

1. **Structure produces behaviour.** If a situation keeps returning to the same state, something in the arrangement is returning it. Finding that something is the work.
2. **The explanation is usually circular, not linear.** A affects B, and B affects A. Causes and effects are not in separate columns, which is why a list of causes rarely explains a persistent problem.
3. **Effects arrive late.** What is done now often shows up years later, and the delay is why people conclude that an intervention failed when it has simply not arrived yet.
4. **Where you draw the edge decides what you can explain.** Draw it too tightly and the cause sits outside it; draw it too widely and nothing can be traced.
5. **Nobody sees the whole system.** Everyone involved sees the part they stand in, and each of those views is reasonable from where that person stands. An analysis that treats one of them as simply wrong has given up on explaining the behaviour.

What it is not

It is not a way of proving what caused something. The output is an explanation good enough to act on, held together by evidence and open to being wrong, and the process keeps saying which parts are evidence and which parts are estimate.

It is also not a way of making a problem bigger. The discipline runs the other way: the edge is drawn deliberately, the number of parts is capped, and everything left out is left out with a reason.

What a system is

A **system** is a set of things, interconnected in such a way that they produce their own pattern of behaviour over time (Meadows, 2008, p. 2).

The three parts

A system has to have all three of these, and the order matters less than the fact that none of them can be missing (Meadows, 2008, p. 11).

PART	WHAT IT MEANS HERE
Elements	The things the system is made of. In this process the elements that matter are the actors, and the quantities that rise and fall
Interconnections	What links the elements: rules, money, information, who depends on whom. This is where most of the explanation turns out to be
A function or purpose	What the arrangement produces, judged by what it actually produces rather than by what anyone says it is for

What is not a system

A collection without interconnections or a function is not a system. Meadows uses sand scattered on a road: add sand or take sand away and there is still just sand on the road. Take a player out of a football team, or a part out of a digestive system, and what remains is a different system (Meadows, 2008, pp. 11 and 12).

The test is useful in the other direction too. If nothing changes when a part is removed, that part is probably not inside the system being analysed.

Two things that follow, and that this workbook relies on

A system is more than the sum of its parts. Listing the parts and describing each one does not explain what the arrangement produces. The interconnections have to be drawn.

The function is read off what is produced, not off what is intended. A well-meant arrangement that reliably produces a result is, for the purposes of analysis, an arrangement that is set up to produce that result. This is uncomfortable and it is the most useful idea here.

Glossary

The short definition of each term, and the page where the term is set out in full. Terms are used precisely in this process and are not swapped for near synonyms.

Actor	Someone who can act in this system, or is acted upon by it. Four tests admit an actor, and any one of them is enough	p. 12
Behaviour	The system's, not anyone's conduct: the path a quantity has taken over time	p. 20
Boundary	The line between what is part of this system and what is next to it but not part of it. Everything outside carries a reason	p. 12
Critical check	The short list at the end of a step, checked by the participant, that has to be complete before the next step starts	p. 24
Desired change	What would be different if the problem improved, said in the same terms as the problem definition and not as a way of getting there	p. 8
Layer	A set of actors who work within the same scope of decision-making or implementation, ordered by how far a decision reaches	p. 12
Pattern shape	The name for what a line on a graph does, chosen from a closed list of six	p. 22
Position	Who the participant is in relation to this system, chosen from five. It decides what the last steps of the process deliver	p. 10
Problem definition	The sentence naming what is wrong with something valued, for whom, since when, and why that matters. Written first as an attempt, then agreed; every later step reads the agreed one	p. 8
Resources	The seven kinds of thing an actor can hold that others in the system need	p. 14
Stakeholder	Not used in this process. Everything inside the system is an actor	p. 12
System	A set of things, interconnected in such a way that they produce their own pattern of behaviour over time	p. 5
System problem definition	One sentence naming the pattern the system keeps producing, which is a different sentence from the agreed problem definition	p. 22

On the platform every one of these terms is a link, so the full definition opens wherever the term appears.

OPPOSITE: PHASE A BEGINS

Frame

Three steps that turn a topic into a system with edges: what the problem is, what is inside the system and whose views count, and how the situation has behaved over time.

STEP 1 · PAGE 9

Define the problem, the desired change, and your position

Fixes what is being analysed, what change is wanted, and who you are in relation to it.

STEP 2 · PAGE 13

Set the boundary and the perspectives

Decides what is inside the system and what is left out, and whose views count.

STEP 3 · PAGE 21

Describe behaviour over time

Shows the pattern the system produces.

Phase A is complete when the critical checks on page 24 can all be ticked off, and what carries forward has been written on page 25.

STEP 1 · BEFORE YOU WRITE

Define the problem

To fix what is being analysed, what change is wanted, and who you are in relation to it, before any explaining starts.

The problem definition

The problem is written as one sentence in a fixed form. That sentence is called the **problem definition** throughout this workbook.

There is **too much** (or **too little**) of **[thing]** for **[who]**, since **[when]**, which matters because **[consequence]**.

Then a short rationale saying how and why people are affected. Where the reason it matters is that something is unfair, the rationale says who is affected, compared with whom, and against which expectation. Two questions are worth answering in the rationale: what is at stake, and why do we care? What is driving us to think about this now?

This form does not fit every problem. Three other forms, and when each is needed, are on page 10.

Eight rules, read against the finished problem definition

1. **One problem per definition.** If the sentence needs "and" to hold two things, there are two problems.
2. **Name a thing that can rise or fall.** Not "the system is failing", which cannot rise or fall, but "the share of applications decided within the legal term", which can.
3. **No solution in the definition.** An action, an instrument, or an organisation that should act is a potential solution: keep it for later, not in the sentence.
4. **No cause in the definition.** "Because the ministry underfunds it" is an answer the analysis has to earn, not a premise to start from.
5. **No blame in the definition.** Rewrite it so that someone who disagrees about the cause could still accept the sentence as a description.
6. **Bound it.** Place, population, and period, so that the questions in Exercise 3 can be answered.
7. **A value can be the reason, not the fact.** "This is unfair" is why it matters. The definition itself still needs what there is too much or too little of.
8. **Plain words.** If a term would need explaining to someone outside the field, define it in the rationale or drop it.

On neutrality

The problem definition is not value neutral and does not pretend to be. Calling something too little or too much implies an expectation about what the level should be. The rule is that the expectation is named in the rationale rather than hidden in an adjective. "Too few inspections" hides the expectation. "Fewer inspections than the twice-yearly minimum the regulation sets" names it.

Define the problem

1

EXERCISE 1

The situation

In your own words, what is going on? A warm-up, not the problem definition.

2

EXERCISE 2

The problem definition, first attempt

There is [] too much [] too little _____
THE THING THERE IS TOO MUCH OR TOO LITTLE OF

for _____ since _____
WHO IT AFFECTS WHEN

which matters because _____
THE CONSEQUENCE

The rationale: how and why people are affected.

3

EXERCISE 3

Is this the kind of problem this process is for?

Y	N		HOW YOU KNOW
☐	☐	Is this an ongoing problem, rather than something that happened once?	_____
☐	☐	Does it have a history you can describe?	_____
☐	☐	Have previous attempts to deal with it failed?	_____
☐	☐	Can you say where the problem starts and stops, and foresee about a dozen things that drive it?	_____

One question disqualifies a topic.

☐ ☐ Could a known procedure solve this, if someone chose to? _____

A yes means the problem is a decision that nobody has taken, not a system that keeps producing a result. Choose a different topic.

4

EXERCISE 4

The desired change

In the same terms as the problem definition in Exercise 2. Not how to get there.

IN FIVE YEARS OR MORE YEAR _____

A NEARER POINT, ONE TO FIVE YEARS OUT YEAR _____

5

EXERCISE 5

Your position

☐ Analysing from outside, with no client and no organisation

Or one of these, each of which acts for or advises someone:

☐ Acting for an organisation or a cause ☐ Advising someone who decides

☐ Deciding, or my organisation does ☐ Affected by this

IF ONE OF THOSE FOUR, ON WHOSE BEHALF? _____

REVISED ON _____ BECAUSE _____

Other forms, positions, and size

The check before Exercise 6

Read the problem definition in Exercise 2 against two questions. Could someone who disagrees about the cause still accept the problem definition as a description? Could that person say what would have to change for the problem definition to stop being true? If either answer is no, the problem definition is carrying a cause, a blame, or a solution, and needs to be rewritten in Exercise 6. If both answers are yes, copy the first attempt across to Exercise 6 as it stands. Either way, the agreed problem definition is always written in Exercise 6, because it is the one every later step reads.

When too much and too little does not fit

The form on page 8 is the default, not the only one. Take these in order, and choose one only with a line saying why the default did not fit. All of them carry the same five parts: the thing, the direction or the gap, who it affects, since when, and why it matters.

FORM	USE IT WHEN, AND HOW TO WRITE IT
Mismatch	The quantity is right, but it is in the wrong place, at the wrong time, or with the wrong people. <i>X exists, but not where (or when, or for whom) it is needed: it is at A while the need is at B.</i>
Falling short of a standard	The thing exists and is counted, but fails a rule, a promise, or a quality that was set. <i>X falls short of Y: the standard says A, the practice is B.</i>
Not happened yet	A risk, or a capacity that does not exist at all. There is no series to be too high or too low. <i>If A continues, B becomes likely, and nothing currently prevents it.</i>

The five positions, and what each one changes

Your position decides what the analysis ends in, so it is chosen once and carried through.

POSITION	WHAT THE ANALYSIS ENDS IN
Analysing from outside	Findings. Your own influence is not part of the analysis
Acting for an organisation or a cause	A target and an ask. Your own influence is part of the analysis
Advising someone who decides	Options with a recommendation. The reach that counts is the client's, not yours
Deciding, or your organisation does	A decision. What is inside your own control is looked at first
Affected by this	A route through an organisation or a coalition, because the power available is mostly power with others

If the problem is too large, or too small

Exercise 3 is how you know. If you cannot say where the problem starts and stops, or cannot foresee about a dozen things that drive it, it is too large. If almost nothing drives it, it is too small.

Narrow by place, by population, by a single decision, or by period. **Widen** by asking why the situation persists, or by taking in the decision that sets the conditions for it. The size question in Exercise 3 is a judgement rather than a rule: it asks whether the problem is small enough to work on and still large enough to be worth explaining.

How others describe it, in Exercise 7

Find how the problem is currently described by those with power over it: a policy document, a press release, a minister's answer, an annual report, a minutes entry. Write their description in their words, not yours, and then write what that description takes for granted and what it leaves out. Give the source. One description is enough here; the others come later.

The agreed problem definition

Almost every first attempt carries a cause, a blame, or a solution. Write the agreed problem definition here, rewritten or copied across, rather than crossing out the first attempt, so that both are visible.

6

EXERCISE 6

The problem definition, agreed

☐ Too much or too little ☐ Mismatch ☐ Falling short of a standard ☐ Not happened yet

IF NOT THE DEFAULT FORM, WHY IT DID NOT FIT _____

THE THING

WHAT IS WRONG WITH IT

WHO IT AFFECTS

SINCE WHEN

WHICH MATTERS BECAUSE

7

EXERCISE 7

How others describe it

In their words. Then what that description takes for granted.

WHERE YOU READ IT _____

WHAT THAT DESCRIPTION LEAVES OUT _____

8

EXERCISE 8

Potential solutions, for later

Potential solutions that occur to you go here, not in the agreed problem definition.

REVISED ON _____ BECAUSE _____

Set the boundary and the perspectives

To decide what is inside the system and what is left out, and whose views count. This is where the case stops being a topic and becomes a system with edges.

Four words this step uses precisely

Actor. Someone who can act in this system or is acted upon by it. Any one of four tests admits an actor: **1** they do something that affects the problem, **2** they are affected by what the system produces, **3** they hold something others in the system need, or **4** they can start or stop a change. Two things are not actors: a category with no agency, such as the weather, the market, or society, which is a condition rather than an actor; and a named individual, unless the thing held belongs to that person rather than to their organisation.

Layer. A set of actors who work within the same scope of decision-making or implementation, meaning that what any of them decides applies to the same territory, population, or part of the organisation. Layers are ordered by how far a decision reaches, from the widest scope of decision to the narrowest, and by nothing else. They are not ranks, not importance, and not a chain of command. They belong to the system, not to you.

Actor type. What kind of actor this is, in this case. The list is written by the participant, at most five types, and as a guide no more than one type per three actors. An actor that seems to need two types usually means the type list is wrong rather than the actor.

Stakeholder is not used in this process. Everything inside the system is an actor.

Drawing the boundary

Almost everything is connected to almost everything else, so "is this related?" admits everything. The test is difference, not relevance: **would including this change what the analysis concludes?** A candidate goes inside if any one of three things is true. The candidate can increase or decrease the quantity named in the agreed problem definition. The candidate is on the receiving end of the excess or deficit of that quantity, meaning the candidate is part of who the problem is for. Or the candidate sets the rules the first two work under.

Everything left out is marked **F** fixed, **O** out of reach, or **I** irrelevant, with one line of reason. Fixed means it will not change during the period being analysed. Out of reach means it could change but nobody in this case can affect it. Irrelevant means including it would not change what the analysis concludes.

The boundary stays provisional. Everything marked F or O is written down rather than forgotten, because a thing assumed fixed that turns out to move is the commonest reason an intervention fails.

How many actors

About a dozen. Past roughly fifteen, either the granularity or the boundary has gone wrong: individual officials are being listed where the department that decides is the actor, or the boundary has taken in layers where nobody passes any of the three tests above. Check granularity first.

The boundary, the layers, and the types

1

EXERCISE 1 · ON A LARGE SHEET, NOT IN THIS BOOK

Draw the system

Put everything you know about this case onto one large sheet. Pictures and symbols rather than sentences, phrases rather than paragraphs, and show who is in conflict with whom. Use marks for money, for conflict, for a blockage, and a question mark for what you do not know. Work standing up, with the sheet flat on a table or on a wall. Nothing on it has to be tidy.

2

EXERCISE 2

What is part of this system, and what is not

List the actors, and anything else that affects the problem: a rule, a flow of money, a procedure. Everything on the right is marked F fixed, O out of reach, or I irrelevant, with a reason.

PART OF THIS SYSTEM	NEXT TO IT, BUT NOT PART OF IT	F	O	I
		☐	☐	☐
	BECAUSE			
		☐	☐	☐
	BECAUSE			
		☐	☐	☐
	BECAUSE			
		☐	☐	☐
	BECAUSE			
		☐	☐	☐
	BECAUSE			
		☐	☐	☐
	BECAUSE			

More rows than these go on the notes pages, from page 26.

3

EXERCISE 3

The layers

Ordered by how far a decision reaches, widest scope of decision first. The narrowest layer goes on row 5, and any layers between on rows 2 to 4. Three to five, and five is the limit. Two only if you write why this case has no third scope of decision, on row 3.

1	WHY THIS IS THE WIDEST SCOPE
2	WHY THIS SCOPE SITS HERE
3	WHY THIS SCOPE SITS HERE
4	WHY THIS SCOPE SITS HERE
5	WHY THIS IS THE NARROWEST SCOPE

4

EXERCISE 4

The actor types

At most five. As a guide, no more than one type per three actors.

A	B	C
D	E	

REVISED ON _____ BECAUSE _____

The resources an actor holds

Power, and why this step records resources instead

Power here is an actor's capacity to affect what the system does: what gets decided, what reaches the agenda, and what goes unquestioned. That capacity is hard to judge directly and easy to guess at. What can be recorded instead is concrete: the **resources** an actor holds that others in the system need. Recording the resources is this step's work; judging what kind of power they add up to comes later, once there is something to judge it against.

The seven resources

Seven kinds, enough to sort by and few enough to remember.

RESOURCE	WHAT IT MEANS
Formal authority	Can decide, or change a rule
Money	Holds or can redirect the funding
Delivery capacity	Does the work on the ground
Information and expertise	Knows something the others need to know
Access	Can get to whoever decides
Legitimacy	Can credibly speak for others
Public attention	Can make an issue visible, or keep it quiet

Two rules while you fill the table

1. An actor that seems to need two layers is usually two actors, as a ministry's policy directorate and its inspectorate are two actors with different reach.
2. An affected group that splits into halves wanting different things is two actors. A single entry that hides the split will not survive the rest of the analysis.

A third rule, about who gets left out

Include those the system acts on, not only those who act. Name the group by its position in the system ("households refused on income grounds since 2023").

The four columns before the resources

Columns **1** to **4** are the four admission tests from page 12: **1** affects the problem, **2** is affected by what the system produces, **3** holds something others need, **4** can start or stop a change. Tick every one that applies. An actor with no tick in any of the four does not belong in the table, and column **2** is the one to read back: if nothing is ticked down that column, the list has only the actors who act, and not those the system acts on.

Who decides, and who disagrees

Authority, responsibility, and accountability are three different things

TERM	THE QUESTION IT ANSWERS	WHERE THE ANSWER USUALLY IS
Authority	Who can take this decision, or change the rule the decision is made under?	A statute, a mandate, a delegation, a budget line someone can redirect
Responsibility	Who is tasked with carrying this decision out?	A work programme, a contract, an implementation plan, a job description
Accountability	Who is called to answer when what was decided goes wrong?	An inspection regime, a reporting duty, an election, a complaints procedure

A layer can hold one without the others, and where the three sit apart that is the finding rather than a complication. It is often the finding: the layer that answers for a failure is frequently not the layer that could have prevented it.

Who the dependency question is asked for

Every actor's resources are recorded in Exercise 5. The second half, who needs each resource from them, is asked only for a subset. The subset is not a matter of your sense of importance: it is read off the allocation in Exercise 6, and completed while you fill Exercise 7. Any actor in a layer holding authority, responsibility, or accountability for the decision, and anyone who holds something those actors need. That is usually three to six actors. Every other actor keeps the ticks in Exercise 5 and nothing more, and that blank is a state rather than an omission.

How the actors see the problem

This is not the question from Step 1 asked twice. Step 1 recorded one description, the one used by those with power over the issue. Here you ask what each actor takes the problem to be, in that actor's own terms, and what that follows from. One line per actor, for the actors whose reading differs from one already written.

Mark each line **evidenced** (a document, a quoted statement, a meeting, an interview) or **inferred**. An inferred line is not guesswork: it names what it is read from. A line that cannot name what it is read from is a guess and stays out. Motive is always a guess, so record what an actor treats as the problem, never why that actor privately wants it.

At least two of the descriptions need to differ in a way that matters, meaning both cannot be acted on at once. Two wordings of the same description do not count. Nothing here asks whether a description is correct.

Who speaks for the groups the system acts on

For each group the system acts on, write the group and who currently speaks for that group on the line under Exercise 9, including "nobody".

The decision, the dependencies, and the views

6

EXERCISE 6
Authority, responsibility, and accountability

THE DECISION THE DESIRED CHANGE DEPENDS ON _____

AUTHORITY _____

RESPONSIBILITY _____

ACCOUNTABILITY _____
"Nobody" is an allowed answer.

7

EXERCISE 7
Who needs what from whom

ACTOR	RESOURCE HELD	WHO NEEDS IT, AND FOR WHAT

More rows than these go on the notes pages, from page 26.

8

EXERCISE 8
How the actors see the problem

ACTOR	WHAT THEY TAKE THE PROBLEM TO BE	EV. / INF.	READ FROM WHAT?

9

EXERCISE 9
Who is not in this list, and why not?

WHO SPEAKS FOR EACH GROUP THE SYSTEM ACTS ON, INCLUDING NOBODY

REVISED ON _____ BECAUSE _____

EXERCISE 10

Draw the system again

Across both pages. The same case, arranged this time by the layers, the actors, and the descriptions you have just defined. Put the actors in bands by layer, widest scope of decision at the top. Draw a line between two actors where one needs a resource from the other, and mark where two actors read the problem differently. Write the three sentences on the right.

EXERCISE 11

Three sentences about what you have drawn

1. What this system produces, whether or not anyone intends it. Read the bands: what does the arrangement reliably hand out, and to whom?

2. Where the disagreement is. Take the two descriptions from Exercise 8 that cannot both be acted on, and say who holds each and what each one wants.

3. What surprised you when you drew it. A missing line, an actor with nothing, a layer that turned out to hold nothing.

REVISED ON _____ BECAUSE _____

Describe behaviour over time

To show the pattern the system produces. This is the first step that looks at the case as something that has been happening rather than as something that is the case.

Behaviour means the system's, not anyone's conduct

It is the path a quantity has taken over time, and nothing about how people acted. Writing about officials being slow is answering a different question.

What to graph

What is measured is fixed. How you measure it is yours. The thing the graph is about is the one the agreed problem definition names: the thing there is too much or too little of. That does not change here, because changing it is how a case quietly changes subject, and nobody notices afterwards because everything that follows is consistent with whatever was graphed.

The measure is a separate choice, and it is yours. One thing can usually be counted several ways, and you take whichever count has a history you can draw: a waiting list, the number of people in emergency accommodation, and nights of shelter used are three measures of one thing. Write down which measure you are using, and why that one. If no measure of the named thing can be found at all, the thing named is not a quantity, which is a return to Step 1 rather than a fault in this step.

If the agreed problem definition used one of the other forms, graph the quantity that the definition compares against something. For a **mismatch**, graph the amount available in the place where the need is. For **falling short of a standard**, graph the measured level of the thing the standard is about. For **not happened yet**, there is no series to draw, so graph the nearest quantity that does have a history: the pressure that would produce the risk, or the capacity that would absorb it, and say why that is the nearest one available.

The period goes back to the last time the system changed

Back to the last major change (a reform, a crisis, a new instrument, a change of government with a policy attached), or roughly ten years, whichever is longer. Not for thoroughness: a period that starts after the last reform shows the system settling down, which looks like stability, and an explanation built for stability explains the wrong thing. Use intervals as coarse as the evidence allows. Yearly is usually enough, and monthly usually produces noise you will then try to explain.

This step tests Step 1

Step 1's second rule was to name a thing that can rise or fall. Here that claim is put to the test.

IF THIS HAPPENS	GO BACK TO
The line cannot be drawn, because the quantity does not move	Step 1, page 11, and rename the thing. What was named is a condition or a judgement, not a quantity
The line can be drawn, but it is not the problem	Step 1, page 11. The problem definition named a proxy
The period has to reach outside your boundary	Step 2, page 13, and reopen the exclusions

This is the commonest return in the whole process, and going back here costs very little.

The pattern

1

EXERCISE 1

What you are graphing

THE THING MEASURED, FROM THE AGREED PROBLEM DEFINITION _____

HOW YOU ARE MEASURING IT, AND WHY THAT MEASURE _____

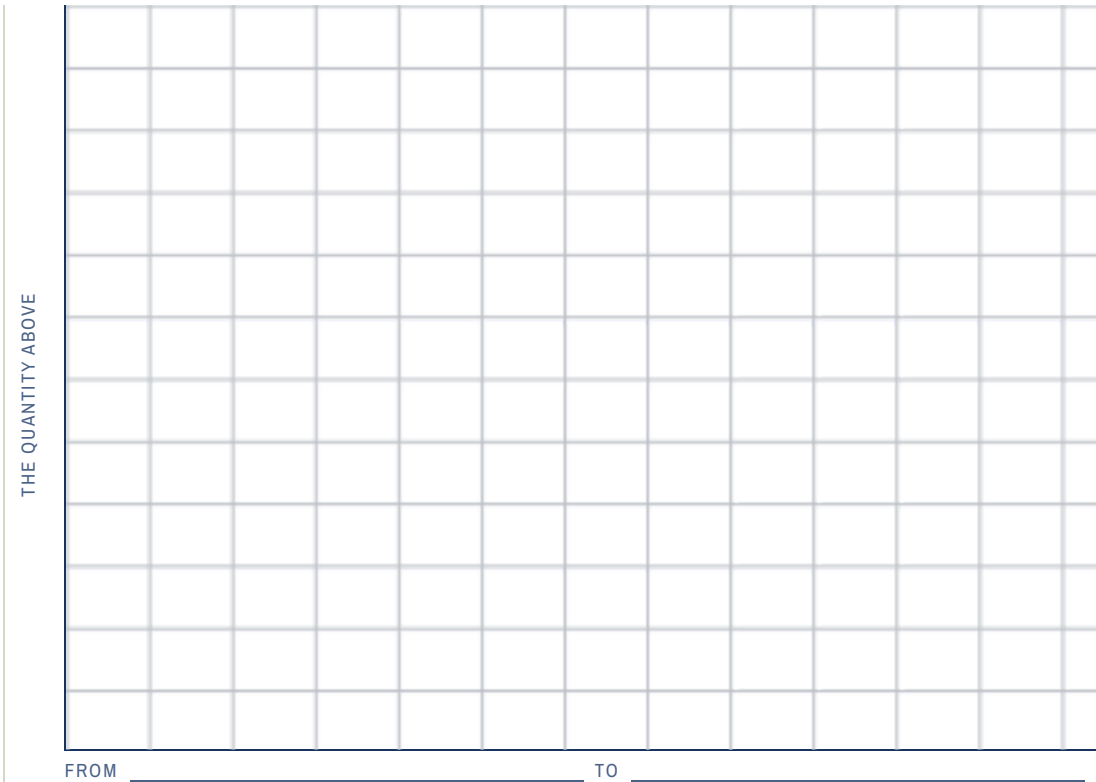
NOT HAPPENED YET: WHY THE QUANTITY GRAPHED IS THE NEAREST _____

THE PERIOD STARTS AT _____ BECAUSE _____

2

EXERCISE 2

The line



3

EXERCISE 3

What happened, and when

Reforms, crises, new instruments, court rulings, and every previous attempt to fix this. Line them up under the years they happened.

--	--	--	--	--	--

Six cells. If there were more, _____ in total, and these are the ones that mattered.

4

EXERCISE 4

Where the line comes from

Mark each part of the line. An unlabelled mixture is the one thing that is not allowed.

☐ Measured, a published series ☐ Documented, from accounts ☐ Estimated, our own reading

REVISED ON _____ BECAUSE _____

Shapes, evidence, and the system problem definition

The six shapes

Pick the one that describes your graph, and then describe why your graph is that shape. A shape you choose is a claim about the system. A shape you describe in your own words is a caption.

SHAPE	WHAT THE LINE DOES
Steady rise or fall	Moves one way at roughly the same rate
Accelerating	Moves one way, faster and faster
Levelling off	Moves one way, then flattens towards a limit
Up and down	Repeatedly overshoots and comes back
Rise then collapse	Grows, turns, and falls below where it started
Flat	Does not move, despite the pressure on it

Flat is a pattern, not a failure. A quantity that stays put while people push on it is being held there by something, and naming what holds it is the work that follows. Most problems that have survived several attempts are this shape.

Evidence is labelled, not ranked

A line drawn from documented accounts is a legitimate graph. What matters is that the reader can tell which kind each part of the line is. Where the line changes kind, mark the change.

KIND	WHAT YOU WRITE BESIDE THE LINE
Measured	The source and the years it covers
Documented	The documents, and what each one establishes about direction or timing
Estimated	What the estimate rests on, and the words "this line is an estimate"

The system problem definition

One sentence naming the pattern. It is a different sentence from the problem definition on page 11: that one says what is wrong, and this one says what the system keeps doing.

Since **[when]**, **[the quantity]** has **[shape]**, and **[what has been tried]** has not changed it.

The two futures

Optional, and drawn on the same axes as the line. One dotted line for what you expect to happen if nothing changes, and one for what the desired change from Step 1 would look like if it happened. Neither is a prediction. Both are drawn before anything has been chosen, which is the point: they fix what better would look like as a shape, while nobody is yet attached to a particular way of getting there.

The shape, and what it means

5

EXERCISE 5

The shape of the line

- ☐ Steady rise or fall
- ☐ Accelerating
- ☐ Levelling off
- ☐ Up and down
- ☐ Rise then collapse
- ☐ Flat

Why your graph is that shape.

6

EXERCISE 6

The system problem definition

Since _____

WHEN THE PERIOD STARTS

_____ has _____

THE QUANTITY THE SHAPE

and _____ has not changed it.

WHAT HAS BEEN TRIED

7

EXERCISE 7

The two futures, and any other quantity

Optional. On the left, a dotted line for what you expect to happen if nothing changes, and a second for what the desired change would look like. Use the right-hand box for one further quantity, where the agreed problem definition rests on it.



REVISED ON _____ BECAUSE _____

BEFORE YOU LEAVE PHASE A

The critical checks

Checked by you. Nobody signs them off. A line that cannot be ticked off sends you back to the page named beside it, and going back is free.

Step 1 is complete when

- ☐ The agreed problem definition states a situation, not a solution: what is wrong with something valued, with the reason it matters and for whom. p. 11
- ☐ Your position is named. p. 9
- ☐ The desired change is stated after the problem, not before it, in the same terms as the agreed problem definition. p. 9
- ☐ The four questions in Exercise 3 are answered, and a known procedure would not solve the problem. p. 9
- ☐ Potential solutions that occurred to you are in Exercise 8, not in the agreed problem definition. p. 11

Step 2 is complete when

- ☐ What is excluded is written down, with a reason and a mark of F, O, or I. p. 13
- ☐ The actors include those affected by the system but not involved in it. p. 15
- ☐ Every actor sits in one layer and carries one type. pp. 13, 15
- ☐ For every actor, the resources held are recorded. For the actors the dependency question applies to, who needs each resource from them is recorded as well. pp. 15, 17
- ☐ Which layer can take the decision, which is tasked with carrying the decision out, and which answers for the result, are each named or recorded as nobody. p. 17
- ☐ At least two descriptions of the problem are recorded, and they differ in a way that matters. p. 17
- ☐ The system has been drawn on a large sheet, drawn again in this book, and its three sentences written. Large sheet, pp. 18, 19

Step 3 is complete when

- ☐ The quantity graphed measures the thing named in the agreed problem definition, and a line says which measure it is and why that one. p. 21
- ☐ The period reaches back to the last major change in the system, or ten years, and a line says which. p. 21
- ☐ The line carries its evidence, with each part marked measured, documented, or estimated. p. 21
- ☐ What has been tried is on the event strip, with the years. p. 21
- ☐ The shape is named, with a line saying why the graph is that shape. p. 23
- ☐ The system problem definition is written, and it is not the agreed problem definition reworded. p. 23

OPPOSITE: WHAT CARRIES FORWARD

PHASE A

What carries forward

The agreed version, copied here. Everything the rest of the process needs from Phase A is on this page, so this is the only page to find again. The page each item was decided on is printed beside it.

1

The problem definition, as agreed

Decided on page 11.

2

The desired change, and your position

Decided on page 9.

3

The boundary, in one sentence, and the exclusion most likely to be wrong

Decided on page 13.

4

The layers, and who holds the decision

Decided on pages 13 and 17. Each layer keeps its row from page 13.

1	
2	
3	
4	
5	

AUTHORITY

RESPONSIBILITY

ACCOUNTABILITY

5

The two descriptions that disagree

Decided on page 17. Name the actor holding each.

6

The three sentences about the drawing

Written on page 19.

7

The system problem definition, and the shape

Written on page 23.

REVISED ON

 BECAUSE

SUMMARY AND NOTES

Notes

Questions you could not answer, sources to find, disagreements the group did not settle, and anything worth keeping that has no working page of its own.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

SUMMARY AND NOTES

Notes

[illegible]

A SYSTEMS THINKING PROCESS FOR POLICY AND INTERVENTION DESIGN

Phase A workbook. Steps 1 to 3 of thirteen. The later phases are added as they are written.

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